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Second Homework

Exercise 1: High Salary Report

Write a PL/SQL program using a cursor to display all employees who earn more than the average salary of their department. Show employee name, salary and department name.

```
SET SERVEROUTPUT ON;
DECLARE
    CURSOR emp_cursor IS
        SELECT e.first_name || ' ' || e.last_name AS employee_name,
            e.salary,
            e.department_name
        FROM employees e
        JOIN departments d ON e.department_id = d.department_id
        WHERE e.salary > (SELECT AVG(salary) FROM employees
        WHERE department_id = e.department_id);
BEGIN
    FOR emp_rec IN emp_cursor LOOP
        DBMS_OUTPUT.PUT_LINE(
            'Employee: ' || emp_rec.employee_name ||
            ', Salary: ' || emp_rec.salary ||
            ', Department: ' || emp_rec.department_name);
    END LOOP;
END;
/
```

Exercise 2: Salary Increment by Department

use a cursor to go through each employee in the IT department and increase their salary by 10%, then display the updated salaries.

```
DECLARE
        CURSOR it_emp_cursor IS
                SELECT employee_id, salary
                FROM employees
                WHERE department_id = 10
                FOR UPDATE OF salary;
        new_salary NUMBER;
BEGIN
        FOR emp_rec IN it_emp_cursor LOOP
                new_salary := emp_rec.salary * 1.10;
                UPDATE employees
                SET salary = new_salary
                WHERE CURRENT OF it_emp_cursor;
                DBMS_OUTPUT.PUT_LINE(
                        'Employee ID: ' || emp_rec.employee_id ||
                        ', New salary: ' || new_salary
                        );
                END LOOP;
                COMMIT;
END;
/
```

Exercise 3: Customer Orders Summary

Write a cursor-based program to list all customers who have placed orders in the last 30 days. For each customer, display the total number of orders and the total order amount.

```
DECLARE
    CURSOR cust_cursor IS
        SELECT c.customer_name,
            COUNT(o.order_id) AS total_orders,
            SUM(o.total_amount) AS total_amount
        FROM customers c
        JOIN orders o ON c.customer_id = o.customer_id
        WHERE o.order_date >= SYSDATE – 30
        GROUP BY c.customer_name;
BEGIN
    FOR cust_rec IN cust_cursor LOOP
        DBMS_OUTPUT.PUT_LINE(
            ‘Customer: ’ || cust_rec.customer_name ||
            ‘, Total Orders: ’ || cust_rec.total_orders ||
            ‘, Total Amount: ’ || cust_rec.total_amount
        );
    END LOOP;
END;
/
```

Exercise 4: Pending Payments Reminder

Write a cursor-based program to list all customers who have placed orders in the last 30 days. For each customer, display the total number of orders and the total order amount.

```
DECLARE
        CURSOR invoice_cursor IS
                SELECT c.customer_name,
                        i.invoice_id,
                        i.due_date,
                        i.amount_due
                FROM invoices i
                JOIN customers c ON i.customer_id = c.customer_id
                WHERE i.status = 'Pending'
                AND i.due_date < SYSDATE - 60;
BEGIN
        FOR inv_rec IN invoice_cursor LOOP
                DBMS_OUTPUT.PUT_LINE(
                        'Customer: ' || inv_rec.customer_name ||
                        ', Invoice ID: ' || inv_rec.invoice_id ||
                        ', Due Date: ' ||
                         TO_CHAR(inv_rec.due_date, 'DD-MON-YYYY') ||
                        ', Amount Due: ' || inv_rec.amount_due
                        );
        END LOOP;
END;
/
```

Exercise 5: Top N Employees per Department

Using a cursor, display the top 3 highest-paid employees in each department. Show department name, employee name, and salary.

```
DECLARE
    CURSOR dept_cursor IS
        SELECT department_id, department_name
        FROM departments;
    CURSOR top_emp_cursor (p_dept_id NUMBER) IS
        SELECT first_name || ' ' || last_name AS employee_name, salary
        FROM employees
        WHERE department_id = p_dept_id
        ORDER BY salary DESC
        FETCH FIRST 3 ROWS ONLY;
BEGIN
    FOR dept_rec IN dept_cursor LOOP
        DBMS_OUTPUT.PUT_LINE(' Department: ' ||
        dept_rec.department_name);
        FOR emp_rec IN top_emp_cursor(dept_rec.department_id) LOOP
            DBMS_OUTPUT.PUT_LINE(
            '   Employee: ' || emp_rec.employee_name ||
            ', Salary: ' || emp_rec.salary
            );
        END LOOP;
    END LOOP
END;
/
```

Exercise 6: Low Stock Alert

Write a cursor to scan the product table and display all products where the stock quantity is less than 10. Show product name, category, and current stock.

```
DECLARE
    CURSOR product_cursor IS
        SELECT product_name, category, stock_quantity
        FROM products
        WHERE stock_quantity < 10;
BEGIN
    FOR prod_rec IN product_cursor LOOP
        DBMS_OUTPUT.PUT_LINE(
        'Product: ' || prod_rec.product_name ||
        ', Category: ' || prod_rec.category ||
        ', Stock: ' || prod_rec.stock_quantity
        );
    END LOOP;
END;
/
```

Exercise 7: Student Performance Report

Using a cursor, list all students who scored below the class average in the latest exam. Show student name, subject, and marks.

```
DECLARE
    CURSOR student_cursor IS
        SELECT student_name, subject, marks
        FROM students
        WHERE marks < (SELECT AVG(marks) FROM students);
BEGIN
    FOR stud_rec IN student_cursor LOOP
        DBMS_OUTPUT.PUT_LINE(
            'Student: ' || stud_rec.student_name ||
            ', Subject: ' || stud_rec.subject ||
            ', Marks: ' || stud_rec.marks
        );
    END LOOP;
END;
/
```

Exercise 8: Employee Anniversary

Write a cursor program to find employees who are completing 5 years, 10 years, or 15 years in the company this month. Display employee name, hire date, and years completed.

```
DECLARE
    CURSOR anniversary_cursor IS
        SELECT first_name || ' ' || last_name AS employee_name,
            hire_date,
            TRUNC(MONTHS_BETWEEN(SYSDATE, hire_date)/12) AS
            Years_completed
        FROM employees
        WHERE EXTRACT(MOTH FROM hire_date) = EXTRACT(MONTH
FROM SYSDATE);
        AND TRUNC(MONTHS_BETWEEN(SYSDATE, hire_date)/12) IN
(5,10, 15);
BEGIN
    FOR emp_rec IN anniversary_cursor LOOP
        DBMS_OUTPUT.PUT_LINE(
            'Employee: ' || emp_rec.employee_name ||
            ', Hire Date: ' || TO_CHAR(emp_rec.hire_date, 'DD-MON-
YYYY') ||
            ', Years Complete: ' || emp_rec.years_completed
        );
    END LOOP;
END;
/
```